

41107 Marine and Ocean Engineering – Formula Sheet

Condensed from the uploaded lecture notes, exam scripts, and practice material.

GENERAL / CONSTANTS

Gravity	$g = 9.81 \text{ m/s}^2$
Water density	$\rho \approx 1025 \text{ kg/m}^3$
Kinematic viscosity	$\nu \approx 10^{-6} \text{ m}^2/\text{s}$
Circle area	$A = \pi(D/2)^2 = \pi D^2/4$
Angular frequency	$\omega = 2\pi/T = 2\pi f$
Frequency	$f = \omega/(2\pi) = 1/T$
Period	$T = 1/f = 2\pi/\omega$
Knots to m/s	$U[\text{m/s}] = 0.51444 U[\text{knots}]$
Wave celerity	$c = L/T = \omega/k$
Wave number	$k = 2\pi/L$

FLOW AROUND CYLINDERS: STEADY CURRENT

Reynolds number	$Re = U_c D / \nu$
Dynamic head	$q = \frac{1}{2} \rho U^2$
Pressure coefficient	$C_p = \frac{p - p_0}{\frac{1}{2} \rho U^2}$
Stagnation pressure	$p_s - p_0 = \frac{1}{2} \rho U^2$
Mean drag	$F_D = \frac{1}{2} \rho C_D D U^2$
Lift	$F_L = \frac{1}{2} \rho C_L D U^2$
Drag coefficient	$C_D = \frac{F_D}{\frac{1}{2} \rho U^2 D}$
Osc. drag coeff.	$C'_D = \frac{F'_D}{\frac{1}{2} \rho U^2 D}$
Osc. lift coeff.	$C'_L = \frac{F'_L}{\frac{1}{2} \rho U^2 D}$
Strouhal number	$St = \frac{f_v D}{U}$
Drag-force frequency	$f_{D,osc} = 2f_v$
Drag dominance	For $Re \gtrsim 10^4$, form drag $\gg$ friction drag.

FLOW AROUND CYLINDERS: OSCILLATORY FLOW

Plane oscillatory flow	$U(t) = U_m \sin(\omega t)$
Wave Reynolds number	$Re = U_m D / \nu$
Keulegan–Carpenter	$KC = \frac{U_m T}{D} = \frac{2\pi a}{D}$
Horizontal excursion	$U_m = 2\pi a / T$
Morison force	$F_x = \frac{1}{2} \rho C_D D U U + \rho C_M A \ddot{U}$
General Morison	$F_x \approx \frac{1}{2} \rho C_D D (U - U_b) U - U_b + \rho c_m A (\ddot{U} - \dot{U}_b) + \rho A \ddot{U}$
Froude–Krylov	$F_{FK} = \rho A \ddot{U}$
Coeff. relation	$C_M = c_m + 1$
For cylinder, pot. flow	$c_m = 1 \Rightarrow C_M = 2$
Inertia/drag ratio	$\frac{F_{I,max}}{F_{D,max}} = \frac{\pi^2}{KC} \frac{C_M}{C_D}$
Rule of thumb	Small KC : inertia-dominated. Large KC : drag-dominated.
Very small KC	Flow may have no separation.
$KC > 7$	Vortex shedding regime appears.

VORTEX-INDUCED VIBRATIONS (VIV)

Basic oscillator	$m\ddot{y} + c\dot{y} + ky = F$
With added mass	$(m + m')\ddot{y} + c\dot{y} + ky = F$
Added mass	$m' = \rho c_m A$
Undamped nat. freq.	$\omega_n = \sqrt{\frac{k}{m + m'}}$
Natural frequency	$f_n = \omega_n / (2\pi)$
Free decay	$y(t) = A_y e^{-\zeta \omega_d t} \cos(\omega_d t)$
Damped nat. freq.	$\omega_d = \omega_n \sqrt{1 - \zeta^2}$
Damping ratio	$\zeta = \frac{2(m + m')\omega_n}{U}$
Free-decay damping	$\zeta \approx \frac{1}{2\pi} \ln \left(\frac{y_n}{y_{n+1}} \right)$
Reduced velocity	$V_r = \frac{U}{D f_n}$
Mass ratio	$M = \frac{m + m'}{\rho D^2} = \frac{m}{\rho D^2} + \frac{\pi}{4} c_m$
Stability parameter	$K_s = 2M(2\pi\zeta_s) = 4\pi M\zeta_s$
Simple amplitude model	$A/D \approx \frac{1.5}{K_s}$
Cross-flow lock-in	Typically around $4 \lesssim V_r \lesssim 8$.
Frequency match	In lock-in: $f \approx f_v \approx f_n$.
Typical amplitudes	Cross-flow A_y usually larger than in-line A_x .

LIFT FREQUENCY IN WAVES

Wave frequency	$f_w = 1/T_w$
Normalized lift freq.	$N_L = f_L / f_w$
Normalized vibration freq.	$N = f / f_w$
Freq. relation	$\frac{f_n}{f_w} = \frac{KC}{V_r}$
Single-pair regime	Typically $N_L \approx 2$.
Double-pair regime	Typically $N_L \approx 3$.
Three-pair regime	Typically $N_L \approx 4$.
Small KC	Usually $f_L \neq f_v$ because of flow reversal.
Large $KC \rightarrow \infty$	Steady-current limit: $f_L \rightarrow f_v$.

DIFFRACTION / LARGE BODIES

Diffraction regime	Important when $D/L \gtrsim O(0.2)$.
Typical region	Large body + waves: often $0 \lesssim KC \lesssim 2$.
Potential flow	$\nabla^2 \phi = 0$
Velocity potential	$u = \phi_x, v = \phi_y, w = \phi_z$
No-flow through body	$\partial \phi / \partial n = 0$
Linear pressure	$p = -\rho \partial \phi / \partial t$
Total potential	$\phi = \phi_i + \phi_s$
Horizontal force	$F_x = - \int_0^{2\pi} p(r_0 d\theta) \cos \theta$
Equivalent inertia form	$F_x = \rho C_M (\pi r_0^2) \dot{u}$
Diffraction inertia coeff.	$C_M = \frac{4A(kr_0)}{\pi(kr_0)^2}$
Small-cylinder limit	$kr_0 \rightarrow 0 \Rightarrow C_M \rightarrow 2$
Froude–Krylov (small cyl.)	$F_{FK} = \rho A \dot{u}$

CORRELATION LENGTH

Correlation coeff.	$R(z) = \frac{p'(\zeta) p'(\zeta + z)}{\sqrt{p'^2(\zeta) p'^2(\zeta + z)}}$
Correlation length	$L_c = \int_0^\infty R(z) dz$
Interpretation	Large L_c : loads fluctuate coherently over a longer span.

REGULAR WAVES: BASIC THEORY

Surface elevation	$\eta(x, t) = \frac{H}{2} \cos(\omega t - kx)$
Amplitude	$\zeta_a = H/2$
Angular frequency	$\omega = 2\pi/T$
Wave number	$k = 2\pi/L$
Celerity	$c = L/T = \omega/k$
Dispersion relation	$\omega^2 = gk \tanh(kh)$
Deep water	$L = \frac{gT^2}{2\pi} = \frac{2\pi g}{\omega^2}$
Horiz. particle vel.	$u = \frac{\pi H}{T} \frac{\cosh(k(z+h))}{\sinh(kh)} \cos(\omega t - kx)$
Vert. particle vel.	$w = \frac{\pi H}{T} \frac{\sinh(kh)}{\cosh(k(z+h))} \sin(\omega t - kx)$
Horiz. motion amplitude	$a = \frac{H}{2} \frac{\cosh(k(z+h))}{\sinh(kh)}$
Surface amplitude	$a_{z=0} = \frac{H}{2 \tanh(kh)}$
Regular-wave energy	$E_{pot} = \frac{1}{4} \rho g \zeta_a^2 L, E_{kin} = \frac{1}{4} \rho g \zeta_a^2 L, E = \frac{1}{2} \rho g \zeta_a^2 L$
Breaking limit	$\frac{H}{L} \lesssim 0.14 \tanh(kh)$
Max height on horiz. bed	$H_{max} \approx 0.14 \tanh(kh) L$

IRREGULAR WAVES / SPECTRA

Wave spectrum	$S_\zeta(\omega_n) \Delta\omega = \frac{1}{2} a_n^2$
Variance	$\sigma^2 = \int_0^\infty S_\zeta(\omega) d\omega = m_0$
Spectral moment	$m_n = \int_0^\infty \omega^n S_\zeta(\omega) d\omega$
Significant height	$H_s = H_1/3 = 4\sqrt{m_0}$
Mean energy period	$T_E = 2\pi m_{-1} / m_0$
Mean period	$T_1 = 2\pi m_0 / m_1$
Zero-upcrossing period	$T_z = 2\pi \sqrt{m_0 / m_2}$
Crest-to-crest period	$T_C = 2\pi \sqrt{m_2 / m_4}$
Peak period	$T_p = 2\pi / \omega_p, \omega_p = \arg \max S(\omega)$
Freq. conversion	$S(\omega) d\omega = S(f) df, \omega = 2\pi f, S(f) = 2\pi S(\omega)$
Bretschneider	$S(\omega) = \frac{A}{\omega^5} \exp\left(-\frac{B}{\omega^4}\right)$
Bretschneider coeffs.	$A = \frac{H_s^2}{4\pi} \left(\frac{2\pi}{T_z}\right)^4, B = \frac{1}{\pi} \left(\frac{2\pi}{T_z}\right)^4$
PM spectrum	$S(\omega) = 8.1 \times 10^{-3} \frac{g^2}{\omega^5} \exp\left[-0.74 \left(\frac{g}{V\omega}\right)^4\right]$
JONSWAP	$S_J(\omega) = \alpha \frac{g^2}{\omega^5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^r$
JONSWAP peak factor	$r = \exp\left[-\frac{(\omega - \omega_p)^2}{2\sigma^2 \omega_p^2}\right]$

PROBABILITY / RAYLEIGH

Surface elevation	Usually modeled as Gaussian with zero mean.
Wave heights / crests	Often modeled as Rayleigh (narrow-banded sea).
Peak-amplitude exceedance	$P(X > x) = \exp\left(-\frac{x^2}{2m_0}\right)$
Crest exceedance	$P(\xi_c > h) = \exp\left(-\frac{h^2}{2m_0}\right)$
Wave-height exceedance	$P(H > h) = \exp\left(-\frac{h^2}{8m_0}\right)$
Expected no. exceedances	Approx. $N_{exc} = N P(\cdot)$, with $N \approx$ no. of waves in the record.

SHIP MOTION DYNAMICS / REGULAR WAVES

6 DOF	Surge, sway, heave, roll, pitch, yaw.
Linear theory	Response proportional to wave amplitude and at same frequency, with phase shift.
Excitation + radiation	$F(t) = F_{exc}(t) + F_{rad}(t) + F_{rest}(t)$
Vertical cylinder EOM	$(m + a)\ddot{z} + b\dot{z} + cz = F_a \cos(\omega t + \epsilon_{F\zeta})$
Restoring coeff. (vert. cyl.)	$c = \rho g A_w = \rho g \pi D^2 / 4$
Undamped nat. freq.	$\omega_n = \sqrt{\frac{c}{m + a}}$
Harmonic response	$z = z_a \cos(\omega t + \epsilon_{z\zeta})$
Velocity	$\dot{z} = -z_a \omega \sin(\omega t + \epsilon_{z\zeta})$
Acceleration	$\ddot{z} = -z_a \omega^2 \cos(\omega t + \epsilon_{z\zeta})$
RAO magnitude	$z_a(\omega) = \Phi(\omega) \zeta_a$
Complex RAO	$\Phi(\omega) = \Re\Phi + i\Im\Phi$
RAO amplitude	$ \Phi = \sqrt{(\Re\Phi)^2 + (\Im\Phi)^2}$
RAO phase	$\epsilon = \text{atan2}(\Im\Phi, \Re\Phi)$
Interpretation	If $\Re < 0$, use the correct quadrant in atan2 (often equivalent to adding 180°).

FORWARD SPEED / ENCOUNTERED WAVES

Encounter freq.	$\omega_e = \omega - \frac{\omega^2 U}{g} \cos \mu$
Encounter period	$T_e = 2\pi / \omega_e$
Energy conservation	$S(\omega) d\omega = S^*(\omega_e) d\omega_e$
Spectrum transform	$S^*(\omega_e) = \frac{1}{1 - 2\omega U \cos \mu / g} S(\omega)$
Angles used in course	Typically beam to head seas: $90^\circ \leq \mu \leq 270^\circ$.

SPECTRAL RESPONSE OF SHIPS / FLOATERS

Wave component energy	$S_\zeta(\omega_n) \Delta\omega = \frac{1}{2} a_n^2$
Response amplitude	$r_0 = \Phi(\omega) a$
Response spectrum	$S_R(\omega) = \Phi(\omega) ^2 S_\zeta(\omega)$
Response moments	$m_{R,n} = \int_0^\infty \omega^n S_R(\omega) d\omega$
Sig. response amplitude	$r_{1/3} = 4 \sqrt{m_{R,0}}$
Peak exceedance	$P(X > x) = \exp\left(-\frac{x^2}{2m_{R,0}}\right)$
Qualitative rule	Large overlap between $S_\zeta(\omega)$ and $ \Phi(\omega) $ gives large response.

SHIP HYDROSTATICS / FLOATATION

Buoyancy = weight	$\rho g \nabla = W$
Box/barge displacement	$\nabla = L B T$
Draught of box/barge	$T = \frac{W}{\rho g L B}$
Weight from cargo only	$W = \rho_s g (L B D_{cargo})$
Center of buoyancy	$KB = T/2$ (rectangular / box-like)
Waterplane inertia	$I_x = \frac{L B^3}{12}$ (transverse)
General I_x	$I_x = \frac{2}{3} \int y^3(x) dx$
Metacentric radius	$B M_T = I_x / \nabla$
Metacentric height from keel	$K M_T = KB + B M_T$
Metacentric height	$G M_T = K M_T - KG$
Righting lever (small heel)	$GZ \approx G M_T \sin \phi$
Restoring moment	$M_R = \rho g \nabla GZ$
Wall-sided formula	$GZ = \sin \phi \left(G M_T + \frac{1}{2} B M_T \tan^2 \phi \right)$
Wall-sided moment	$M_R = \rho g \nabla \sin \phi \left(G M_T + \frac{1}{2} B M_T \tan^2 \phi \right)$
Barge with M_T in WL	$K M_T = T \Rightarrow T = B / \sqrt{6}$
Stability criterion	Initial stability if $G M_T > 0$; unstable if $G M_T < 0$.

SHIP RESISTANCE / SCALING

Froude number	$F_n = \frac{U}{\sqrt{gL}}$
Reynolds number	$Re = \frac{U L}{\nu}$
ITTC friction coeff.	$C_F = \frac{0.075}{(\log_{10} Re - 2)^2}$
Friction resistance	$R_F = \frac{1}{2} \rho U^2 S C_F$
Total resistance	$R_T \approx (1 + k) R_F + R_W$
Effective power	$P_E = R_T U$
Geometric scale	$\alpha = L_s / L_m$
Area scaling	$S_s / S_m = \alpha^2$
Volume scaling	$\nabla_s / \nabla_m = \alpha^3$
Froude speed scaling	$\frac{U_s}{U_m} = \sqrt{\alpha}$
Froude time scaling	$\frac{T_s}{T_m} = \sqrt{\alpha}$
Mass/weight scaling	$\frac{W_s}{W_m} \approx \alpha^3$
Resistance scaling (approx.)	$\frac{R_s}{R_m} \approx \alpha^3$ at corresponding speeds
Power scaling (approx.)	$\frac{P_s}{P_m} \approx \alpha^{3.5}$
Model testing	In practice, preserve Froude similarity ; exact Reynolds similarity is not possible simultaneously.

STRENGTH OF SHIPS / BENDING

Distributed load	$q(x) = b(x) - w(x)$
Still-water load	Buoyancy upward minus weight downward.
Shear-force relation	$\frac{dV}{dx} = -q(x)$
Bending-moment relation	$\frac{dM}{dx} = V(x)$
Neutral axis	$z_n = \frac{\sum A_i z_i}{\sum A_i}$
Second moment of area	$I = \int_A z^2 dA$
Steiner / parallel-axis	$I = \sum \left(I_i + A_i a_i^2 \right)$
Thin-walled approx.	$I \approx \sum b_i t_i z_i^2$
Navier bending stress	$\sigma_{11} = \frac{M y z}{I}$
Section modulus	$Z = \frac{I}{z_{max}}$
Max bending stress	$\sigma_{max} = \frac{M}{Z}$
Stress location	Largest $ \sigma $ occurs furthest from the neutral axis.
Hogging / sagging	Wave or still-water load distribution sets the sign of bending moment.

WAVE-INDUCED BENDING / SHORT-TERM STATS

VBM response spectrum	$S_M(\omega_e) = \Phi_M(\omega_e) ^2 S_e(\omega_e)$
VBM response moments	$m_{R,n} = \int_0^\infty \omega_e^n S_M(\omega_e) d\omega_e$
Rayleigh exceedance	$P(X > x) = \exp\left(-\frac{x^2}{2m_{R,0}}\right)$
Applies to	Short-term prediction in a stationary sea state.

VERY COMMON EXAM SHORTCUTS

Stagnation pressure	Use $C_p = 1$ at the stagnation point.
Oscillating drag freq.	Use $2f_v$.
Lift regime name	If $N_L = 3$, think double-pair regime.
Small log/barge stability	For a floating log / box, compute $G M_T = KB + B M_T - KG$.
Response from spectrum	Area under response spectrum $\Rightarrow m_{R,0} \Rightarrow$ significant amplitude.
Encountered waves	Use ω_e first, then convert back to T_e .
KC on seabed	Find local orbital amplitude a at $z = -h$, then use $KC = 2\pi a / D$.